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Like or Not to Like: An Usecase of Vietnamese Street Food Videos on YouTube

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PDF Download
3746027.3755864.pdf
28 March 2026
Total Citations: 0
Total Downloads: 128

Published: 27 October 2025

Citation in BibTeX format

MM '25: The 33rd ACM International Conference on Multimedia
October 27 - 31, 2025
Dublin, Ireland

Conference Sponsors:
SIGMM

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Abstract

Vietnamese street food has gained global popularity through platforms like YouTube, where creators are incentivized to post videos on specific topics, including street food. Efficient video editing has become essential for YouTubers. This study analyzed 1,507 videos and gathered insights from 213 respondents to identify the factors that shape viewer preferences and highlight key video features for creators. The study's key finding was the strong synergy between visual elements and linguistic richness, emphasizing the connection between storytelling and what appears on screen. Using machine learning, we examined visual, linguistic, and acoustic features, achieving a 70.5% accurate predictive model. Compared to the baseline 50%, these insights led to personalized recommendations for street food content creators, offering strategies to enhance viewer engagement. Our automated recommendation system bridges data-driven insights with content creation, elevating Vietnamese street food on the global stage and celebrating the blend of local culture and technology.

CCS Concepts

• **Human-centered computing → User studies**.

Keywords

StreetFood, AI for Good, Youtube, Machine Learning

ACM Reference Format:
Duy X. Nguyen, Hoan V. Hoang, Ninh A. Vu, Loc T. Nguyen, and Trung T. Phan. 2025. Like or Not to Like: An Usecase of Vietnamese Street Food Videos on YouTube. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3746027.3755864

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MM '25, Dublin, Ireland
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ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3755864

1 Introduction

Street food, consisting of ready-to-eat foods sold by vendors in public spaces, is a vital part of urban food consumption globally, valued for its affordability, convenience, and reflection of local culinary traditions [3, 17, 23, 58]. Despite food safety concerns, its vibrant flavors and cultural authenticity make it integral to urban life [15, 29, 30]. Street food supports local economies, attracts tourists, and sustains livelihoods, especially in developing countries, and vendors contribute significantly to job creation and the informal economy [42, 51]. However, they face challenges such as health regulations and competition with formal food sectors, hindering their growth [4, 14]. In Vietnam, street food is not just a culinary tradition but a key part of the nation's cultural identity, with dishes like Pho and Banh Mi offering an authentic culinary experience that attracts both locals and tourists. These iconic foods contribute to Vietnam's food tourism industry, positioning the country as a premier destination for food tourism [19, 31, 38, 43].

Traveling is closely linked with culinary experiences, and street food often takes center stage in this connection. Food tourism has become a significant trend, with travelers seeking authentic culinary experiences to immerse in local culture [22, 43]. Street food offers an affordable and genuine way to experience local traditions, providing a unique taste of place. Tourists often document their street food adventures through photos, videos, and written narratives, capturing the dynamic aspects of food culture [22, 41, 45]. Platforms like YouTube and TikTok are popular for sharing these experiences, with videos preferred for their ability to capture movement, sounds, and interactions, enhancing viewer engagement [2, 57]. Vlogging has enabled travelers to share high-quality content, attracting large audiences and potential monetization through brand partnerships [16]. To effectively share their culinary journeys, travelers need skills in video editing, understanding trends, and managing multimedia. Machine learning algorithms in video editing can improve content quality and help creators optimize videos for better reach and impact [67].

In Vietnam, YouTube's vast reach and popularity make it a significant medium for content creators to connect with audiences and promote culinary culture [9, 21]. This paper targets to understand streetfood in Vietnam through lens of YouTubers. To the best of our knowledge, there is no research about understanding Vietnamese street food by using social media data, i.e., YouTube videos. Hence, the research gap is to understand street food and propose methods

to infer the possibility of street food videos to get liked by viewers on YouTube.

To solve the research gap, we raise the following two research questions:

- **RQ1**: Given a collection of street food videos in Vietnam from YouTubers, what key patterns in visual and textual content can be revealed through video features?
- **RQ2**: Given extracted visual-textual content and annotated features that identify meaningful factors in street food videos in Vietnam, can machine-learning models be applied to grade the videos, then to guide creators in improving future video designs tailored for this genre?

The contributions of the paper are the following:

1. Our research examines 1,507 Vietnamese street food videos from YouTube to explore social media's role in global cultural exchange. Focusing on YouTube's influence in sharing cultural content, we analyze how Vietnamese cuisine is portrayed and consumed worldwide. This study offers insights into viewer preferences, engagement, and the effectiveness of digital media strategies in promoting food culture.

2. In RQ1, our study identifies key factors in Vietnamese street food videos through automated feature extraction using a pre-trained deep learning model. This analysis uncovered common objects, locations, scenes, and audio elements reflecting Vietnamese street food culture. Narration styles were analyzed with LIWC, revealing a use of future-oriented language at the start and end of videos, often tied to viewer engagement. Viewer impressions highlighted storytelling and recording styles as significant influencers, with correlations showing that external sounds negatively impact perceptions, while transcript word count, food mentions, and emotional content enhance engagement. We also developed a shot segmentation method based on cosine similarity, averaging 127 shots per video at 6.7 seconds each. These findings deepen understanding of video content's role in engaging audiences within this cultural context.

3. In RQ2, we utilized a machine learning model to classify Vietnamese street food videos and predict viewer preferences based on extracted features. Combining object features with linguistic elements from LIWC, the random forest model achieved the highest accuracy at 70.5%. The prominence of LIWC features highlights the vital role of storytelling in engaging viewers. This model provides actionable insights for content creators, guiding video tailoring to audience preferences and optimizing elements such as objects, places, scenes, audio, and shot composition. These findings support marketers and creators in enhancing video visibility and engagement, promoting Vietnamese culture, and fostering commercial success by aligning content with viewer expectations.

The paper is organized as follows. Related work discusses previous work. Dataset describes the dataset and the cleaning process. RQ1 presents the audio-visual content analysis of street food videos. RQ2 presents the video classification tasks and results. Discussion discusses the findings and limitations. Conclusion concludes the paper.

2 Related work

2.1 Street Food and Enhancing Food Culture Through Photos and Videos

Street food shapes urban life by creating vibrant public spaces that foster social interactions, attract tourists, and strengthen community bonds, contributing to urban cultural dynamics [1, 55, 60]. Economically, it provides employment, supports entrepreneurship in the informal sector, and boosts local economies [1, 33]. Street food trends reflect socio-economic changes, with vendors adapting to consumer preferences and global influences, highlighting the dynamic nature of urban societies [11, 17]. This study shifts focus to the digital representation of Vietnamese street food on platforms like YouTube. Unlike traditional research on physical street food spaces, it examines how storytelling, visuals, and audio in online videos shape global perceptions. The analysis explores how digital media preserves cultural identity while adapting to global trends, extending street food culture beyond local markets. Machine learning models analyzing video content and viewer responses offer novel insights into the evolving relationship between street food and its global audience.

Photos of street food capture its vibrant colors, textures, and contexts, showcasing global food cultures. These visuals transcend cultural barriers, enabling worldwide audiences to appreciate diverse culinary traditions without firsthand experience [56]. Photography also documents culinary practices, promotes cultural identity, and encourages culinary tourism through social media and digital platforms [7]. In this study, we analyzed video frames to highlight street food culture and related characteristics. Videos offer a dynamic representation by incorporating visual, auditory, and interactive elements. They capture food preparation, vendor skills, and ambiance, providing an immersive experience that fosters deeper appreciation of culinary traditions [20, 50]. This medium preserves cultural heritage and enhances engagement by showcasing the craftsmanship and social context behind dishes. Our research uses videos as primary input to investigate these aspects further.

2.2 Impact of YouTube on Food Culture

Social media platforms like YouTube allow global sharing and interaction with content, making it a powerful tool for communication, education, and entertainment. By promoting cultural exchange, YouTube enables creators to share culinary skills, recipes, and food experiences, fostering appreciation through comments and likes [36, 52]. The platform transforms food culture by hosting cooking tutorials, culinary documentaries, and food vlogs, democratizing content creation and distribution while preserving endangered cuisines and promoting culinary diversity. By showcasing preparation techniques and cultural stories, YouTube fosters culinary tourism and deepens appreciation for food as cultural expression [32, 68]. Effective food-related YouTube videos must be visually appealing and educational, using storytelling, high-quality visuals, and clear audio to engage audiences. Incorporating machine learning tools can optimize content for viewer preferences and improve engagement analysis, making content creation more efficient [59, 63]. This study explores video features influencing the likelihood of a video being liked.

2.3 Machine Learning in Video Content Creation

Machine learning (ML) has revolutionized digital content creation, particularly in food videos, by enhancing visual aesthetics, content relevance, and viewer engagement. By analyzing large datasets, ML can identify critical success factors like visual appeal, narrative structure, and emotional resonance [5]. Key applications of ML in video content creation include procedural content generation through generative adversarial networks (GANs), which automate tasks and create unique scenes, similar to techniques used in video games like Doom and Super Mario [27, 44]. Additionally, ML models are used for automated editing tasks, such as scene detection and highlight reel creation, allowing editors to focus on creative aspects [18]. AI video generators like Synthesia also automate video production with AI avatars, speeding up the creation of marketing and educational content [39, 64].

3 Dataset

3.1 Data preparation

We employed the **youtubesearchpython** [54] module to collect data on global perspectives of Vietnamese street food by searching YouTube for video IDs using keywords such as "street food in Vietnam" and "Vietnamese street food", etc. To enhance diversity, the keyword set was expanded to include cities and tourist destinations known for their vibrant street food scenes, like Hanoi, Saigon, Da Lat, etc. Duplicate videos were systematically removed, and two key filters were applied: ensuring English transcript availability to facilitate analysis and verifying that each video was filmed in Vietnam and prominently featured food. Collaborative efforts among team members further refined the dataset, resulting in an initial collection of 1,608 videos.

Throughout the evaluation, challenges included 9 videos becoming private or unavailable and 92 flagged as age-restricted due to inappropriate content. Using the *pytube* library [24], we downloaded a total of 1,599 videos, eventually curating a final dataset of 1,507 videos. This meticulous process ensured the dataset accurately aligned with our research objectives, offering a robust foundation for analyzing how Vietnamese street food is represented and consumed globally. The number of videos retained at each stage is detailed in Table 1.

No.	Stages	Total videos (N)
1	Using 22 keywords to search for potential videos on YouTube	2,871
2	Applying constraints	1,608
3	Removing unavailable videos after annotating	1,599
4	Removing videos with age constraints	1,507

Table 1: Total number of videos after each phase

3.2 Annotation

Building on the curated dataset from the Data Preparation phase, we implemented an online survey to collect data on YouTube viewers' perceptions, preferences, and engagement with Vietnamese street food videos. Online surveys effectively reach diverse populations, minimizing biases and achieving higher response rates [25, 37]. The Google Forms survey, distributed via email to regular YouTube food viewers [10], explored viewer behaviors, attitudes, and preferences related to Vietnamese street food videos. It consisted of three sections: demographic information, YouTube food review-watching habits, and ratings of five Vietnamese street food videos based on music, voice, sound, storytelling, visuals, and shooting style using a five-point Likert scale [47]. These elements were chosen for their role in shaping viewer engagement and emotional connection [6, 48, 62, 66].

3.3 Annotation Responses

To ensure broad applicability and minimize gender bias, we selected 213 annotators, with 95% from ages 18-23, a key demographic for YouTube users. Gender distribution was nearly equal, with 53% male and 47% female, reducing the likelihood of gender-based bias. Additionally, 76% of participants reported watching food review videos multiple times per week, indicating high familiarity and interest in the subject. The demographic composition of the annotators, predominantly Gen Z (ages 18-23) with balanced gender representation, strengthens the credibility and impartiality of our study's results.

The majority of features found in our study were unanimously agreed upon by the annotators, indicating a shared perspective on the main factors that impact audience decisions. When examining street food films, 73.15% of annotators agreed that the food's composition was a key factor. The recording style also played a significant role, with 70.37% acknowledging its impact on the visual narrative. Visual content, showcasing vibrant street scenes and cultural nuances, was important for 58.33% of participants. Environmental noise, common in street food films, negatively affected the viewing experience. Additionally, 62.96% of annotators noted the importance of storytelling for engagement. However, elements like stall decorations, people in the videos, video length, and other objects had less influence, with inter-annotator agreements below 40%.

3.4 Annotation Validation

This section evaluates the quality and reliability of our datasets to ensure the validity of our findings. We used the Intraclass Correlation Coefficient (ICC), specifically ICC(2, k), as recommended by [35], to assess the reliability of annotations from three annotators. This metric is suitable for cases where a fixed group of judges evaluates each target in this case, the dataset of 1,507 videos. The ICC(2, k) indicates the level of consensus among annotators. Following [35], we interpreted ICC scores as follows: values below 0.5 indicate poor reliability, scores between 0.5 and 0.75 suggest moderate reliability, 0.75 to 0.9 indicate high reliability, and scores above 0.9 reflect excellent reliability. It is noteworthy that ICC(2, k) applies only to numerical variables, excluding categorical data.

The viewer preference ICC of Like or Not Like (Question 8 in Table 2) was 0.69, indicating moderate to high agreement with a mean score of 3.53, suggesting generally positive reception. This shows that YouTubers have effective methods for selecting and presenting content. Future research should focus on factors like voice, narrative style, visuals, and recording techniques to refine "likability" ratings and guide future video content analysis. As a

No.	Our questions	ICC (2, k)	Mean	STD	Skew
1	Music in videos	**0.51**	3.52	0.59	-0.026
2	Voice of the YouTuber(s) appearing in the videos?	**0.56**	3.70	0.62	-0.248
3	Environmental sounds in the videos?	0.41	3.11	0.67	0.393
4	Storytelling style of the videos?	**0.63**	3.59	0.75	-0.304
5	Visual information of the videos?	**0.57**	3.68	0.75	-0.417
6	Video style of the videos?	**0.65**	3.41	0.77	0.07
7	Objects shown in the videos?	0.4	3.68	0.54	-0.233
8	Do you like this video?	**0.69**	3.53	0.73	-0.249

Table 2: Street food video factor ICC with N(videos)=1,507, N(raters)=3 and scale (1-5)

short summary, we obtain 1,507 videos which are annotated. We will use this dataset in this work.

4 Vietnam Street Food Patterns Analysis (RQ1)

4.1 Visual content

Visual aesthetics (Visuals of the Videos (Questions 5 to 7 in Table 2)), including image quality and shooting techniques, significantly influenced viewer attraction. ICC values of 0.57 for image quality and 0.65 for shooting style reflected YouTubers' focus on these aspects. Shooting technique scored 3.41/5, and visual information scored 3.68, highlighting high picture quality. However, the ICC for objects in tracking shots was lower at 0.4, likely due to unclear survey instructions and the spontaneous nature of street food videos, emphasizing the need for clearer guidelines in future research

To provide a detailed understanding of the visual content in each video, we employed a deep-learning model to assess the likelihood of objects appearing in each frame. This method gives us a comprehensive perspective on the various elements present. The Inception-V3 model [61], trained on the ImageNet Large Visual Recognition Challenge dataset [53], consisting of 1,000 object categories, was used. However, as this dataset includes items from global contexts, some objects may not align with the Vietnamese setting. The model calculates the probability of an object belonging to any of the 1,000 classes for each frame, with the total probability always summing to 1.

Our analysis in the list of recognized objects uncovered a significant abundance of cooking utensils, including plates, pots, soup bowls, and woks, along with common Vietnamese establishments like restaurants and grocery stores. Notably, objects like "burrito" were frequently identified, visually resembling Vietnamese dishes like spring rolls and Banh Mi. However, there were instances where certain establishments were mistakenly labeled as "toyshops" possibly due to the variety of products displayed on shelves, a common sight in Vietnamese restaurants. For each frame, we obtain a 1,000-dimensional vector, where each element represents the probability of a specific object. We then consolidate these probabilities across all frames within each video, calculating the average probability for each class. This aggregation process is conducted at the video level, ensuring a comprehensive account of all identified items.

To capture the YouTuber's location, we used the Resnet18 model [69], trained on the Place365-Standard dataset [69], which includes 365 location categories such as kitchens, streets, and restaurants. The model assigns probabilities to each category, ensuring total coverage of all outcomes. Commonly predicted locations, like "restaurant" and "street" align with food consumption contexts. Misclassifications sometimes occur, especially with Vietnamese streets, where food displayed in glass cabinets may lead to mislabeling as "ice cream parlors". This reflects similarities in how street food establishments are organized. The frequent appearance of "beauty salon" in videos suggests that content creators use specific filming techniques, such as close-ups and visually appealing settings, to engage viewers. To accurately depict the geolocation, we averaged vectors across all frames, ensuring comprehensive spatial context in street food videos.

To better understand the environmental backdrop of YouTubers, we utilized a deep-learning model with a Resnet150 dilated encoder and a ppm_deepup decoder [71], trained on the MIT SceneParse150 dataset from ADE20K [70]. We calculated the coverage percentage of each scene element in every frame to analyze common visuals in street food videos, revealing frequent depictions of sky, walls, and people. Notably, "floor; flooring" and "ceiling" values appeared often, indicating distinct filming styles. High camera angles highlight ceilings, while low angles emphasize floors. By averaging values across frames, we identified the most prevalent scene characteristics, enhancing our understanding of the visual narratives in street food videos.

4.2 Narration content

Auditory elements such as the presenter's tone (ICC 0.56) and narrative style (ICC 0.63) demonstrated reasonable reliability, with an average emotional score of 3.7/5 for vocal quality and 3.59 for storytelling style, highlighting effective engagement. Despite occasional disruptions from ambient street sounds [46], YouTubers generally maintained good sound quality, emphasizing its importance. However, audio quality showed inconsistencies, with ICCs of 0.51 for music and 0.41 for environmental sound, partly due to unclear survey questions. After excluding 496 videos with less than 5% music content using the YAMNET model [28], the ICC for music dropped further to 0.43, reflecting disagreements in evaluation. Variability in environmental noise also caused low agreement, suggesting future surveys should refine questions and focus on noise levels for better clarity.

To capture a comprehensive audio representation at the scene level, we utilized the YAMNET model [28] to extract 521 distinct audio classes [26]. Notably, the original Audioset data release [26] included 527 classes, but six categories, such as "gendered versions of Speech and Singing, Battle cry, and Funny music," were excluded to prevent objectionable mislabeling based on fairness reviewer recommendations [28]. Trained on the AudioSet dataset, YAMNET excels in detecting common audio events, particularly urban sounds, by analyzing 0.96-second audio samples and generating scores for each class. In street food videos, the most frequent audio categories are "Speech" and "Music". This prevalence reflects the conversational nature of these videos, where sharing personal stories is

common, while music enhances the ambiance. This aligns with the model's sensitivity to audio features commonly found in urban and narrative settings [28].

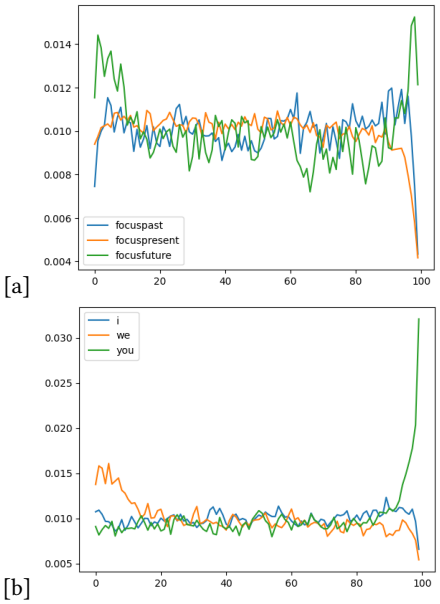

Figure 1: LIWC2022 classes in timeline. [a] Time orientation and [b] Pronouns

Using the Linguistic Inquiry and Word Count (LIWC2022) [12], we analyzed YouTubers' video transcripts to categorize language variables such as Tone, Authenticity, pronoun usage (e.g., I, We, You), and psychological processes like focuspast, focuspresent, and focusfuture. This analysis provided insights into language patterns, emotional states, and cognitive processes. Figure 1 highlights the use of future-oriented language ("focusfuture"), peaking early when discussing plans and again at the end when offering recommendations and engaging viewers, fostering a sense of community and sustaining interest. Pronoun usage also evolves throughout videos, with "We" dominating introductions as YouTubers outline plans and shifting to personal narration mid-video. Toward the end, "You" increases as YouTubers offer recommendations and thank viewers, enhancing engagement and connection.

4.3 Correlation between Video Features

4.3.1 Visual Content and Narration Content In evaluating viewer enjoyment of street food videos, our analysis, shown in Table 3, identifies key factors. The YouTuber's voice and recording style have significant impacts, with correlation coefficients of 0.71. VI is closely linked to RS, with a strong positive association of 0.78, indicating how visual elements influence storytelling. SS and RS collectively affect viewer pleasure, with a correlation of 0.87. To replicate the YouTuber's narrative method, we will analyze both auditory and written content, considering objects, locations, and environments to study filmmaking styles and factors affecting video appreciation. We utilized deep learning techniques to extract features from 1,507 videos, including temporal, frame-based, and video shot segmentation features. Frames were sampled every second, and a deep learning model extracted object, location, and scene features. For audio features, we used the moviepy [72] tool and a deep learning model to analyze 50%-overlapping audio samples of 0.96 seconds. Textual analysis combined dialogues into a string, analyzed using LIWC [12] to reveal distinctive features.

	V	SS	VI	RS	**Like or not to like**
Voice (V)	1	0.78	0.64	0.71	0.78
Storytelling style (SS)	0.78	1	0.73	0.80	0.87
Visual information (VI)	0.64	0.73	1	0.78	0.82
Recording style (RS)	0.71	0.80	0.78	1	0.87

Table 3: Pearson correlation between factors ($p < 0.05$)

4.3.2 Automatic features and viewer impression The Pearson correlation analysis between Like or not to like score and automatic features limited to classes highlights key narration features influencing video likability. Reducing external noise, such as "Engine" (-0.21), "Noise" (-0.22), and "Outside, urban or manmade" (-0.25), is essential to enhance commentary clarity and overall audio-visual quality, improving viewer reception.

The "WC" (Word Count) metric, with a correlation of 0.31, reflects audience engagement through narration. Videos with higher speaking rates (4 words/s) scored better than slower ones (2 words/s), suggesting fast-paced narration attracts larger audiences. This aligns with [65], which found faster-paced videos garner more views.

Specific storytelling elements also influence viewer preferences. Positive correlations were noted for "food" mentions (0.22), expressions of "feeling" (0.22), and "Analytic" evaluations (0.27). A cheerful "Tone" (0.29) enhances attractiveness, encouraging audience engagement and likes. These insights can guide creators in crafting engaging narratives.

4.4 Proposed method for shot segmentation

4.4.1 Shot definition In video analysis, shot segmentation is essential for breaking down a video into segments, each representing a continuous action or event from a single perspective. Traditionally, shot segmentation relied on descriptors like Speeded Up Robust Features (SURF) and Hue Saturation Value (HSV) histograms [8] to assess frame similarity within a shot. Color histograms and video compression techniques, such as Discrete Cosine Transform (DCT) coefficients [34], also aid in detecting shot boundaries by identifying changes in color patterns and compression features. The proposed approach enhances shot segmentation by employing deep learning to extract attributes from each frame, including objects, locations, scenes, and audio. This method builds on previous techniques, integrating advanced deep learning for improved accuracy and efficiency in capturing the recording and narrating style of content.

4.4.2 Proposed shot segmentation algorithm We propose an approach that combines visual and audio characteristics for video shot segmentation. This process partitions a video into distinct shots, crucial for accurate analysis. Our algorithm generates a 2,036-dimensional vector for each frame, incorporating visual elements

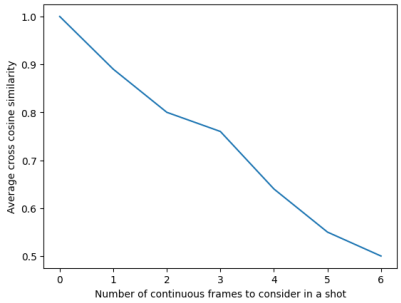

Figure 2: The average cross-cosine similarity of frames within a shot depends on its length. This measurement begins at the frame at one second and concludes at the pre-defined shot length.

(objects, sceneries) and audio features. Given that shots typically have consistent camera viewpoints with minor scene changes, we use cosine similarity to assess frame similarity within a shot. By calculating the cosine of the angle between non-zero vectors, we determine whether frames belong to the same shot. A similarity of 1 indicates identical frames, while 0 suggests they belong to different shots.

Examining the average cosine similarity between frames within the same shot reveals a trend where similarity decreases as more frames are included, reflecting the dynamic nature of videos, such as changes in camera angles and scene transitions (Figure 2). To establish a cosine threshold for frame similarity, each video is divided into ten segments, with the average cosine similarity computed for frames in each segment. This process is repeated across all videos, yielding a cumulative threshold of 0.575 (Figure 3). Our shot segmentation algorithm, integrating visual and audio features, enhances video analysis by improving adaptability and accuracy for tasks like editing and scene identification.

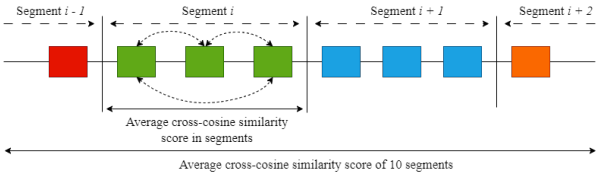

Figure 3: Calulating threshold for shot boundary detection

4.4.3 Shot segmentation results Using the cosine similarity threshold, we segmented 1,507 videos, each averaging 12.4 minutes, into shot boundaries. On average, each video contains about 127 shots, with the highest having 2,198 shots, highlighting varied shot durations and compositions. The average shot duration is approximately 6.7 seconds. This segmentation provides insights into the temporal structure of the videos and lays a foundation for more detailed analysis. We use descriptive statistics to derive features such as maximum and minimum shot durations, duration percentiles, and overall shot counts, offering a quantitative overview of shot distribution and quality. These metrics, when applied to viewer preference and engagement, help calculate mean values for visual and narration characteristics across all shots, providing a comprehensive view of each video. This approach improves our understanding of video content and enhances predictions related to viewer engagement by analyzing the video as a whole.

5 Classifying and recommendation models (RQ2)

Understanding video features is crucial, not only for analysis but also for enhancing video quality and appeal. Focusing on Vietnamese street food, our system aims to assist content creators by recommending improvements in voice modulation, music, narration style, and visual presentation. Looking ahead, we envision a platform where users actively upload street food videos, receiving personalized suggestions to improve voice, music, narration, and visuals. The goal is to help creators produce content that resonates with audiences while continuously refining their skills.

5.1 Classification Models

Amidst various modeling techniques, we selected Random Forest (RF) [13], Linear Regression (LR), and K-Nearest Neighbors (KNN) for viewer sentiment prediction. For RF, we optimized parameters like 'n_estimators', 'max_features', 'max_leaves', and 'criterion' to balance performance and prevent overfitting. In LR, we adjusted the C' parameter to control regularization, with higher values fitting the data more closely and lower values preventing overfitting. For KNN, we tuned 'n_neighbors' to balance complexity and accuracy. Using FLAML, we aimed to optimize these models' performance.

5.2 Evaluation metrics

We evaluate our model using various metrics to assess its performance. Accuracy measures overall correct predictions but can be misleading in imbalanced datasets. F1 Score combines precision and recall, offering a balanced view of handling positive predictions and identifying positive instances, especially when false positives/negatives are costly. Average F1 Score averages across all classes, showing performance adaptability. Uplift measures improvement over a baseline, gauging advancement. These metrics together provide a comprehensive evaluation of the model's strengths and weaknesses.

5.3 Experiment Configurations

For our experiments, we utilized Azure Databricks and Azure as our Cloud Service Provider. The setup featured 10 servers, each with 64 CPU cores and 32 GB of RAM. Azure Databricks, running Spark, allowed us to scale our cluster from 3 to 10 worker nodes, each equipped with 250 GB of RAM and 64 CPU cores. Data was stored in Azure Blob storage using the parquet format, ensuring efficient data management. This infrastructure supported our computational and analytical needs, providing a solid foundation for our experiment results.

5.4 Experiment results

At the core of our endeavors lies the quest to determine whether viewers harbor positive or negative sentiments toward a video. Through a binary classification approach, viewers are asked a straightforward question during the annotation procedure: "Do you like this video or not?" This question forms the basis for our sentiment categorization. The average score from three annotators

becomes the defining metric, with videos scoring below 3 labeled as "disliked" (assigned 0), and those scoring 3 or higher marked as "liked" (assigned 1). Our dataset unfolds with 260 videos (17.3%) in the disliked category and 1,247 (82.3%) in the liked category.

5.4.1 Result Comparison The imbalance in the dataset prompted a deeper analysis and the implementation of random undersampling to address class imbalance in our deception detection dataset. This technique reduces positive class instances (truthful videos) to match the smaller negative class (deceptive videos). Additionally, we explored oversampling, which duplicates minority-class instances to enhance their representation [40, 49]. Both methods aim to balance the dataset using different approaches. Following undersampling and oversampling, the adjusted baseline accuracy increased to 0.5, reflecting the accuracy of a model predicting based solely on the majority class.

In both methods of upsampling and downsampling, improved performance was observed across all feature classes, with LIWC displaying the highest increase of 17.9-18.8% on oversampling and downsampling, respectively. Object and scene follow at 13-16%. These results were yielded by using RF. In other words, RF gives better results compared to LR and KNN. Moreover, frame-based features (F) give better results than shot-based features (S).

The evaluation of resampling strategies, including undersampling and oversampling, showed a significant improvement in average uplift by 17.9-18.8%. While the accuracy gains were similar (18.8% for undersampling and 17.9% for oversampling), oversampling preserved all positive data points, making it preferable for use cases requiring minimal information loss. This factor guided our choice of oversampling as the preferred technique. Among all models, Random Forest consistently outperformed others, with accuracy improvements 14-15% higher than KNN and 13-15% higher than Logistic Regression. The Frame aggregation approach (F) also demonstrated the greatest improvement over Shot-based (S) aggregations. To summarize, the recommended methods for conducting additional analysis involve utilizing a combination of visual and narrative data. The suggested model is Random Forest, the preferred resampling approach is oversampling, and the recommended aggregation method is frame-based aggregations.

5.4.2 Individual and Combined Features In the advanced stages of our study, we aim to combine features from various sources, including LIWC (L), Object1000 (O), Place365 (P), Audio521 (A), Scene150 (S150), and Shot Meta (SM). This integration will provide a comprehensive analytical framework, enhancing the precision and predictive power of our study. The results in Table 4 highlight the best feature combinations, with LIWC and Object1000 consistently showing significant impact on performance, regardless of the feature count. These two features are crucial for predicting viewer preferences for Vietnamese street food videos.

When evaluated alone, LIWC markers outperformed other features, achieving an accuracy of 0.679, indicating the significant role of the transcript in shaping viewer judgments. Combining LIWC and Object1000 boosted accuracy to 0.705, showing the synergistic effect of text and visuals. This suggests that emotional framing and visual context play a key role in how viewers interpret and react to video content, where food, audio, storytelling, and recording style were identified as key factors.

No.	Feature(s)	Acc	F1 Avg	F1 Neg	F1 Pos
1	L	0,679	0,678	0,667	0,689
2	L, O	**0,705**	**0,705**	**0,695**	**0,714**
3	L, O, P	0,700	0,699	0,689	0,709
4	L, O, P, A	0,702	0,701	0,687	0,715
5	L, O, P, A, SM	0,700	0,700	0,690	0,709
6	L, O, P, A, SM, S150	0,684	0,683	0,667	0,698

Table 4: Result of combinations for each count of features within the model. In model column: L (LIWC), O (Object1000), P (Places365), A (Audio521), SM (Shot_meta), S150 (Scene150), F1 Avg (F1 Score Average), F1 Neg (F1 Score Negative), F1 Pos (F1 Score Positive). Note that the Baseline for each case is 0,5.

In conclusion, while linguistic markers are key predictors of preferences, combining textual and visual inputs creates a balanced representation, where each modality compensates for the other's weaknesses. Linguistic structures give meaning to objects, while tangible evidence supports verbal claims. Further research with advanced modeling can explore the interaction between these modalities. However, adding more features does not always improve accuracy; it may even decrease performance and robustness. Thus, feature selection must be careful, considering the relevance and impact of each feature to ensure a model that is both accurate and resilient.

5.4.3 Important Features Our experiments with algorithms such as Random Forest, linear regression, and K-nearest neighbor revealed that the oversampled Random Forest model performed best. This model examines automatic features, including visual and narrative elements. Visual features capture items, places, and scenarios in videos, revealing audience preferences for engaging physical environments and aesthetics. Narrative elements, such as audio quality and LIWC2022 features, offer insights into emotional, cognitive, and structural aspects of narration. Although we extracted the top 30 influential traits, space constraints prevent their display. Combining these features significantly enhances the model's ability to predict viewer preferences, providing valuable guidance for optimizing video content to increase engagement.

5.5 Visual features Importance

The random forest model predominantly uses the top 30 food-related classes, such as soup bowls, zucchinis, ice lollies, etc. Interestingly, animals like ringneck snakes and hermit crabs also appeared, highlighting the variety in Vietnamese street food. The visual resemblance between ringneck snakes and chicken wings, and hermit crabs with snails, emphasizes how appearance shapes the perception of food. Location features include outdoor areas like "athletic_field/outdoor," representing open spaces in Vietnamese street food films. Scenes with locals, such as motorbike rides, underscore the vibrant street culture. Architectural categories like "mansion" and "airfield" further reflect the urban setting. The top 30 visual elements indicate that human characters positively influence viewers. The "airplane" class often refers to culinary equipment, while items like "orange cake," a deep-fried Vietnamese dessert, are identified as the "ball" class, linking street food to the audience.

5.6 Narration features Importance

In the top feature importance score of each class for narrative feature, sound significantly affects how viewers perceive and engage with media. After identifying the most influential audio components, we analyzed the top 100 videos containing these features. Videos with disruptive sounds, like loud traffic or passing trains, were poorly received, likely due to inadequate sound editing. One frequent issue was the "watercraft" sound, caused by YouTubers speaking too closely to the microphone, creating a noise similar to ocean waves or a motorboat. The "Medium engine" sound, a crackling noise due to poor recording quality, also negatively impacted the experience. In contrast, background music had a positive effect on viewer perception, reinforcing the idea that well-chosen music enhances video quality. In analyzing LIWC2022 categories, words expressing superficial emotions ("wow", "sooo", etc.) and satisfaction ("enough", "full", "complete") ranked highly. Expressing emotions and genuine feelings during filming significantly boosts viewer engagement.

5.7 Shot segments Importance

Key insights from the analysis highlight the importance of average shot duration and the number of shots in a video for predicting viewer preference. Videos with "not liked" labels often have long, right-skewed shot durations exceeding 20 seconds, suggesting that extended focus on a single object or scene can lead to viewer disinterest. Similarly, videos with fewer shots lack visual diversity and appear dull, while those with too many shots risk seeming chaotic and unprofessional. The optimal range, around 120 shots per video, provides a balanced and engaging experience, offering clear guidance for content creators.

Our findings from RQ2 revealed critical factors shaping viewer preferences in Vietnamese street food videos. The Random Forest model, enhanced by oversampling, effectively combined visual and narrative data. Linguistic features such as word count, sentence length, and punctuation from the LIWC dataset were particularly influential, emphasizing the role of narration and language in driving engagement. These results underline the importance of thoughtful content preparation, where both visuals and language play pivotal roles. This understanding will inform future models, offering deeper insights into the interplay of linguistic and visual elements in content creation.

6 Discussion

6.1 Understanding Street Food of Vietnam through lens of YouTube

This study marks a pioneering exploration of Vietnamese street food content on YouTube, offering the first systematic inquiry into this previously uncharted area. By focusing on content curated by non-Vietnamese YouTubers, our research provides a unique perspective on how Vietnam's vibrant street food culture is depicted globally. Beyond academic significance, this work serves as a cornerstone for cultural exchange, fostering a deeper global appreciation for Vietnamese street food while potentially driving local economic growth through heightened international awareness. The research also addresses ethical considerations and economic opportunities for stakeholders. Global exposure places greater responsibility on street food vendors to maintain hygiene standards while urging content creators to protect local vendors' livelihoods and cultural authenticity. Additionally, young Vietnamese annotators engaged with the study gained deeper insights into their local street food culture, further emphasizing the societal impact of this work. This dual focus on cultural sensitivity and economic potential positions our research as both an academic and practical contribution to understanding and promoting Vietnam's culinary heritage.

6.2 Potentiality for Automatic Recommendation System in Video Analysis

Our study contributes to video recommendations, benefiting filmmakers, educators, and the artistic community. Using data analysis, crowdsourcing, and collective intelligence, we provide insights that bridge machine learning and the arts, enriching the filmmaking process with actionable data-driven strategies. The recommendation system developed enhances viewer engagement and helps creators refine content. By analyzing shot segmentation and the interaction of visual and linguistic elements (e.g., Object1000, LIWC features), our research reveals key storytelling and aesthetic strategies. This framework advances video analysis and improves content creation, fostering innovation in digital media production.

6.3 Limitations and Future Directions

This study focuses on Vietnamese street food content on YouTube, with potential applications to other culinary and cultural contexts. Future research should apply this methodology to diverse cuisines, such as Thai or Swiss, to enhance generalizability and expand investigative scope. Exploring these variations can deepen insights into cultural nuances and audience preferences globally. The findings underscore the complexity of audience engagement, highlighting the need for further research into psychological factors influencing viewer interactions. Such studies could improve recommendation systems by aligning content with diverse audience tastes. By integrating cultural exploration, ethical considerations, and machine learning, this paper lays a foundation for a more culturally sensitive and informed digital landscape.

7 Conclusion

This study combined manual annotation, multimedia feature extraction, and predictive modeling to analyze Vietnamese street food videos on YouTube. Addressing RQ1, we identified engagement patterns and preferences by integrating crowdsourced judgments with algorithmic analysis of linguistic, visual, narrative, and audio-visual signals. Multi-annotator coding of 1507 videos revealed semantic and visual factors as dominant resonance drivers, further supported by multimedia descriptors of objects, scenes, and sounds. For RQ2, we demonstrated the potential of computational models to predict video reception and guide creators. Our Random Forest approach, leveraging textual and visual features, effectively forecasted video likes while offering personalized recommendations for optimization. Despite dataset limitations in some classification areas, our pipeline outperformed random baselines, showcasing its utility for researchers studying online cultural diffusion.

References

[1] 2017. *Food Trucks, Cultural Identity, and Social Justice.* doi:10.7551/mitpress/9780262036573.001.0001

[2] 2021. A scientist's guide to social media: YouTube, Instagram, and TikTok. https://spie.org/news/photonics-focus/novdec-2021/a-scientists-guide-to-social-media--youtube-instagram-tiktok#_=_

[3] n.d.. Food for the Cities: Street foods. https://www.fao.org/fcit/food-processing/street-foods/en/. https://www.fao.org/fcit/food-processing/street-foods/en/ Food and Agriculture Organization of the United Nations.

[4] Michele Abbott and Deborah Cohen. 2017. What Street Food Vendors in Uganda Need to Succeed. https://www.rand.org/pubs/commentary/2017/05/what-street-food-vendors-in-uganda-need-to-succeed.html

[5] J.K. Aggarwal and Ryoo. 2011. Human activity analysis: A review. *Comput. Surveys* 43, 3 (April 2011), 1–43. doi:10.1145/1922649.1922653

[6] Uchechukwu C Ajiwe and Samuel O Chukwu-Okoronkwo. 2015. Perception of sound: a study of selected Nollywood video films. *Journal of Social Sciences and Humanities* 1, 2 (2015), 48–58.

[7] Roxanne Amerson and Wade Livingston. 2014. Reflexive Photography: An Alternative Method for Documenting the Learning Process of Cultural Competence. *Journal of transcultural nursing : official journal of the Transcultural Nursing Society / Transcultural Nursing Society* 25 (02 2014). doi:10.1177/1043659613515719

[8] Evlampios Apostolidis and Vasileios Mezaris. 2014. Fast shot segmentation combining global and local visual descriptors. *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (2014). doi:10.1109/icassp.2014.6854873

[9] Sara Atske and Sara Atske. 2024. Social Media Use in 2021. https://www.pewresearch.org/internet/2021/04/07/social-media-use-in-2021/

[10] Amol K Bhalerao. 2015. Application and performance of Google Forms for online data collection and analysis: A case of North Eastern Region of India. *Indian Journal of Extension Education* 51, 3and4 (2015), 49–53.

[11] Sharit K. Bhowmik. 2005. Street vendors in Asia: A review. https://api.semanticscholar.org/CorpusID:168829912

[12] Ryan Boyd, Ashwini Ashokkumar, Sarah Seraj, and James Pennebaker. 2022. The Development and Psychometric Properties of LIWC-22. (02 2022). doi:10.13140/RG.2.2.23890.43205

[13] Leo Breiman. 2001. *Machine Learning* 45, 1 (2001), 5–32. doi:10.1023/a:1010933404324

[14] Ray Bromley. 2000. Street vending and public policy: a global review. *International Journal of Sociology and Social Policy* 20, 1/2 (Feb. 2000), 1–28. doi:10.1108/01443330010789052

[15] Joana Campos, Joana Gil, Joana Mourão, Luísa Peixe, and Patrícia Antunes. 2015. Ready-to-eat street-vended food as a potential vehicle of bacterial pathogens and antimicrobial resistance: An exploratory study in Porto region, Portugal. *International Journal of Food Microbiology* 206 (2015), 1–6. doi:10.1016/j.ijfoodmicro.2015.04.016

[16] Wenyi Cheng, Ruwen Tian, and Dickson K.W. Chiu. 2023. Travel vlogs influencing tourist decisions: information preferences and gender differences. *Aslib Journal of Information Management* 76, 1 (Jan. 2023), 86–103. doi:10.1108/ajim-05-2022-0261

[17] Ian Cook and Philip Crang. 1996. The world on a plate: culinary culture, displacement and geographical knowledges. *Journal of material culture* 1, 2 (1996), 131–153.

[18] Paul Covington, Jay Adams, and Emre Sargin. 2016. Deep Neural Networks for YouTube Recommendations. In *Proceedings of the 10th ACM Conference on Recommender Systems* (Boston, Massachusetts, USA) *(RecSys '16)*. Association for Computing Machinery, New York, NY, USA, 191–198. doi:10.1145/2959100.2959190

[19] Nguyen Thi Bich Dao. 2019. A Perception into Food Image and Revisit Intention for Local Cuisine from Foreign Tourist Perspective – The Case of Ho Chi Minh City – Vietnam. *European Journal of Business Management and Research* 4, 2 (May 2019). doi:10.24018/ejbmr.2019.4.2.40

[20] Alessia D'Andrea and Arianna D'Ulizia. 2023. Preserving Local Food Traditions: A Hybrid Participatory Approach for Stimulating Transgenerational Dialogue. *Societies* 13, 4 (2023), 1–15. https://EconPapers.repec.org/RePEc:gam:jsoctx:v:13:y:2023:i:4:p:95-:d:1115415

[21] Frank Elberzhager, Patrick Mennig, and Phil Stüpfert. 2022. Building a YouTube Channel for Science Communication. In *Database and Expert Systems Applications - DEXA 2022 Workshops*, Gabriele Kotsis, A. Min Tjoa, Ismail Khalil, Bernhard Moser, Alfred Taudes, Atif Mashkoor, Johannes Sametinger, Jorge Martinez-Gil, Florian Sobieczky, Lukas Fischer, Rudolf Ramler, Maqbool Khan, and Gerald Czech (Eds.). Springer International Publishing, Cham, 179–188.

[22] Ashleigh Ellis, Eerang Park, Sangkyun Kim, and Ian Yeoman. 2018. What is food tourism? *Tourism Management* 68 (2018), 250–263. doi:10.1016/j.tourman.2018.03.025

[23] Peter Fellows and Martin Hilmi. 2011. Selling street and snack foods. https://api.semanticscholar.org/CorpusID:211799503 FAO Diversification booklet series.

[24] Nick Ficano, Harold Martin, and Taylor Fox Dahlin. 2021. Pytube/pytube: A lightweight, dependency-free Python Library (and command-line utility) for downloading YouTube videos. https://github.com/pytube/pytube.git

[25] Ronald D Fricker and Matthias Schonlau. 2002. Advantages and disadvantages of Internet research surveys: Evidence from the literature. *Field methods* 14, 4 (2002), 347–367.

[26] Jort F. Gemmeke, Daniel P. W. Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R. Channing Moore, Manoj Plakal, and Marvin Ritter. 2017. Audio Set: An ontology and human-labeled dataset for audio events. In *Proc. IEEE ICASSP 2017*. New Orleans, LA.

[27] Edoardo Giacomello, Pier Luca Lanzi, and Daniele Loiacono. 2018. DOOM Level Generation Using Generative Adversarial Networks. In *2018 IEEE Games, Entertainment, Media Conference (GEM)*. 316–323. doi:10.1109/GEM.2018.8516539

[28] Shawn Hershey, Sourish Chaudhuri, Daniel P. W. Ellis, Jort F. Gemmeke, Aren Jansen, R. Channing Moore, Manoj Plakal, Devin Platt, Rif A. Saurous, Bryan Seybold, Malcolm Slaney, Ron J. Weiss, and Kevin Wilson. 2017. CNN Architectures for Large-Scale Audio Classification. arXiv:1609.09430 [cs.SD]

[29] Jillian Hill, Zandile Mchiza, Thandi Puoane, and Nelia P. Steyn. 2019. The development of an evidence-based street food vending model within a socioecological framework: A guide for African countries. *PLOS ONE* 14, 10 (2019). doi:10.1371/journal.pone.0223535

[30] Sinh Hoang and Zuzana Tučková. 2021. The impact of sensory marketing on street food for the return of international visitors: Case study in Vietnam. *Scientific Papers of the University of Pardubice. Series D, Faculty of Economics and Administration* 29, 2 (July 2021). doi:10.46585/sp29021282

[31] Annette Miae Kim. 2015. *Sidewalk City: Remapping Public Space in Ho Chi Minh City*. University of Chicago Press, Chicago. doi:10.7208/9780226119366

[32] Hyun Ji Kim. 2024. An exploratory content analysis of comments and thumbnails of YouTube mukbang videos using computational methods. https://repositories.lib.utexas.edu/items/02c07948-23c4-467d-b048-482b7195a260

[33] A. J. Knox, H. Bressers, N. Mohlakoana, and J. De Groot. 2019. Aspirations to grow: when micro- and informal enterprises in the street food sector speak for themselves. *Journal of Global Entrepreneurship Research* 9, 1 (May 2019). doi:10.1186/s40497-019-0161-7

[34] Vikrant Kobla, David S. Doermann, and King-Ip Lin. 1996. Archiving, indexing, and retrieval of video in the compressed domain. *SPIE Proceedings* (1996). doi:10.1117/12.257312

[35] Terry K. Koo and Mae Y. Li. 2016. A guideline of selecting and reporting intraclass correlation coefficients for Reliability Research. *Journal of Chiropractic Medicine* 15, 2 (2016), 155–163. doi:10.1016/j.jcm.2016.02.012

[36] Mark Lashley. 2015. Making Culture on YouTube: Case Studies of Cultural Production on the Popular Web Platform. *Lasalle* (May 2015). https://www.academia.edu/12338656/Making_Culture_on_YouTube_Case_Studies_of_Cultural_Production_on_the_Popular_Web_Platform

[37] Marija Topuzovska Latkovikj and Mirjana Borota Popovska. 2019. Online research about online research: advantages and disadvantages. *E-methodology* 6, 6 (2019), 44–56.

[38] Kuan-Huei Lee and Noel Scott. 2014. Food Tourism Reviewed Using the Paradigm Funnel Approach. *Journal of Culinary Science Technology* 13, 2 (Dec. 2014), 95–115. doi:10.1080/15428052.2014.952480

[39] Daniel Leiker, Ashley Ricker Gyllen, Ismail Eldesouky, and Mutlu Cukurova. 2023. Generative AI for Learning: Investigating the Potential of Learning Videos with Synthetic Virtual Instructors. In *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium and Blue Sky*, Ning Wang, Genaro Rebolledo-Mendez, Vania Dimitrova, Noboru Matsuda, and Olga C. Santos (Eds.). Springer Nature Switzerland, Cham, 523–529.

[40] Junnan Li, Qingsheng Zhu, Quanwang Wu, and Zhu Fan. 2021. A novel oversampling technique for class-imbalanced learning based on SMOTE and natural neighbors. *Information Sciences* 565 (2021), 438–455. doi:10.1016/j.ins.2021.03.041

[41] Stephen Litvin, Ronald Goldsmith, and Bing Pan. 2008. Electronic Word-of-Mouth in Hospitality and Tourism Management. *Tourism Management* 29 (06 2008), 458–468. doi:10.1016/j.tourman.2007.05.011

[42] Lihua Ma, Hong Chen, Huizhe Yan, Lifeng Wu, and Wenbin Zhang. 2019. Food safety knowledge, attitudes, and behavior of street food vendors and consumers in Handan, a third tier city in China. *BMC Public Health* 19, 1 (Aug. 2019). doi:10.1186/s12889-019-7475-9

[43] Athena H.N. Mak, Margaret Lumbers, and Anita Eves. 2012. Globalisation and food consumption in tourism. *Annals of Tourism Research* 39, 1 (2012), 171–196. doi:10.1016/j.annals.2011.05.010

[44] Xinyu Mao, Wanli Yu, Kazunori D Yamada, and Michael R. Zielewski. 2024. Procedural Content Generation via Generative Artificial Intelligence. arXiv:2407.09013 [cs.AI] https://arxiv.org/abs/2407.09013

[45] Aurelio Mauri and Roberta Minazzi. 2013. Web Reviews Influence on Expectations and Purchasing Intentions of Hotel Potential Customers. *International Journal of Hospitality Management* 34 (05 2013), 99–107. doi:10.1016/j.ijhm.2013.02.012

[46] Tara P McAlexander, Robyn RM Gershon, and Richard L Neitzel. 2015. Street-level noise in an urban setting: Assessment and contribution to personal exposure. *Environmental Health* 14, 1 (2015). doi:10.1186/s12940-015-0006-y

[47] Alexander McLeod, Sonja Pippin, and Jeffrey A Wong. 2011. Revisiting the Likert scale: can the fast form approach improve survey research? *International Journal of Behavioural Accounting and Finance* 2, 3-4 (2011), 310–327.

[48] Jeanne Mengis, Davide Nicolini, and Mara Gorli. 2018. The video production of space: How different recording practices matter. *Organizational research methods*

21, 2 (2018), 288–315.

[49] Roweida Mohammed, Jumanah Rawashdeh, and Malak Abdullah. 2020. Machine Learning with Oversampling and Undersampling Techniques: Overview Study and Experimental Results. In *2020 11th International Conference on Information and Communication Systems (ICICS)*. 243–248. doi:10.1109/ICICS49469.2020.239556

[50] Nikolaos Partarakis, Danae Kaplanidi, Paraskevi Doulgeraki, Effie Karuzaki, Argyro Petraki, Daniele Metilli, Valentina Bartalesi, Ilia Adami, Carlo Meghini, and Xenophon Zabulis. 2021. Representation and Presentation of Culinary Tradition as Cultural Heritage. *Heritage* 4, 2 (2021), 612–640. doi:10.3390/heritage4020036

[51] Thilde Rheinländer, Mette Olsen, John Abubakar Bakang, Harriet Takyi, Flemming Konradsen, and Helle Samuelsen. 2008. Keeping Up Appearances: Perceptions of Street Food Safety in Urban Kumasi, Ghana. *Journal of Urban Health* 85, 6 (Sept. 2008), 952–964. doi:10.1007/s11524-008-9318-3

[52] Gitanjali Roy. 2020. Authenticity v/s Glocalization as Represented in the Digital Platforms: A Study on the Food Culture with Special Reference to Tripura. *Rupkatha Journal on Interdisciplinary Studies in Humanities* 12 (10 2020). doi:10.21659/rupkatha.v12n5.rioc1s17n7

[53] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander Berg, and Li Fei-Fei. 2014. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision* 115 (09 2014). doi:10.1007/s11263-015-0816-y

[54] Hitesh Kumar Saini and Mitja Ševerkar. 2022. YouTube-Search-Python: search for YouTube videos, Channels playlists. https://github.com/alexmercerind/youtube-search-python.git

[55] Caroline Skinner Sally Roever. 2016. Street vendors and cities. *Environment and Urbanization* 28, 2 (2016), 359–374. doi:10.1177/0956247816653898 arXiv:https://doi.org/10.1177/0956247816653898

[56] Joan M. Schwartz and James R. Ryan. 2021. *Picturing Place.* doi:10.4324/9781003268260

[57] Aliaksandra Shutsko. 2020. *User-Generated Short Video Content in Social Media. A Case Study of TikTok.* 108–125. doi:10.1007/978-3-030-49576-3_8

[58] Tulasi Srinivas. 2006. AS MOTHER MADE IT': THE COSMOPOLITAN INDIAN FAMILY,AUTHENTIC'FOOD AND THE CONSTRUCTION OF CULTURAL UTOPIA. *International Journal of Sociology of the Family* (2006), 191–221.

[59] Aswin Punathambekar Sriram Mohan. 2019. Localizing YouTube: Language, cultural regions, and digital platforms. *International Journal of Cultural Studies* 22, 3 (2019), 317–333. doi:10.1177/1367877918794681 arXiv:https://doi.org/10.1177/1367877918794681

[60] Nelia Patricia Steyn, Zandile Mchiza, Jillian Hill, Yul Derek Davids, Irma Venter, Enid Hinrichsen, Maretha Opperman, Julien Rumbelow, and Peter Jacobs. 2013. Nutritional contribution of street foods to the diet of people in developing countries: a systematic review. *Public Health Nutrition* 17, 6 (May 2013), 1363–1374. doi:10.1017/s1368980013001158

[61] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. 2016. Rethinking the inception architecture for computer vision. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2016). doi:10.1109/cvpr.2016.308

[62] Naoya Takahashi, Michael Gygli, and Luc Van Gool. 2017. Aenet: Learning deep audio features for video analysis. *IEEE Transactions on Multimedia* 20, 3 (2017), 513–524.

[63] Joseph M. Terantino. 2011. YouTube for foreign languages: You have to see this video. *Language Learning Technology* 15, 1 (Feb. 2011), 10. https://digitalcommons.kennesaw.edu/cgi/viewcontent.cgi?article=2616&context=facpubs

[64] Carmen Vallis, Stephanie Wilson, Daniel Gozman, and John Buchanan. 2023. Student Perceptions of AI-Generated Avatars in Teaching Business Ethics: We Might not be Impressed. *Postdigital Science and Education* 6, 2 (June 2023), 537–555. doi:10.1007/s42438-023-00407-7

[65] Dustin J. Welbourne and Will J. Grant. 2015. Science communication on YouTube: Factors that affect channel and video popularity. *Public Understanding of Science* 25, 6 (2015), 706–718. doi:10.1177/0963662515572068

[66] Todd Wilkens, Anthony Hughes, Barbara M Wildemuth, and Gary Marchionini. 2003. The role of narrative in understanding digital video: An exploratory analysis. *Proceedings of the American Society for Information Science and Technology* 40, 1 (2003), 323–329.

[67] Jiaheng Xie and Xiao Liu. 2022. Unbox the Blackbox: Predict and Interpret YouTube Viewership Using Deep Learning. arXiv:2101.01076 [cs.LG] https://arxiv.org/abs/2101.01076

[68] Michael K. Goodman Zeena Feldman. 2021. Digital food culture, power and everyday life. *European Journal of Cultural Studies* 24, 6 (2021), 1227–1242. doi:10.1177/13675494211055501 arXiv:https://doi.org/10.1177/13675494211055501

[69] Bolei Zhou, Agata Lapedriza, Aditya Khosla, Aude Oliva, and Antonio Torralba. 2017. Places: A 10 million Image Database for Scene Recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2017).

[70] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. 2017. Scene Parsing through ADE20K Dataset. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.*

[71] Bolei Zhou, Hang Zhao, Xavier Puig, Tete Xiao, Sanja Fidler, Adela Barriuso, and Antonio Torralba. 2018. Semantic understanding of scenes through the ade20k dataset. *International Journal on Computer Vision* (2018).

[72] Zulko. [n. d.]. Zulko/moviepy: Video editing with python. https://github.com/Zulko/moviepy